



Lecture notes - ECO601

Principles Of Economics (Torrens University Australia)



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- **What is Economics?** - Economics is the study of how people make decisions in the face of **scarcity of resources** to satisfy **unlimited wants**. These can be individual decisions, family decisions, business decisions or societal decisions.
- **Problem of Scarcity** - A good is scarce if the choice of one alternative requires that another be given up.
Because of scarcity, most of us never have enough income to buy all the things we want.
- **Key questions for decision making** - What are the alternative uses the resource could be put to?
Which alternative use provides the greatest return?
What would be the net gain for the agent of the current choice relative to the best alternative use?
- **Scarcity, choice, and opportunity cost** - The existence of alternative uses forces us to make choices.
Economic agents, producers and consumers alike are forced to make trade-offs. That is, having more of one good thing means having to do with less of another in the face of resource scarcity.
- **Opportunity cost** - The opportunity cost of any choice is the value of the best alternative forgone in making it.
A fundamental principle of economics is that every choice has an opportunity cost.
Opportunity costs underpins all economic decisions.
- **Example of OC** - If you spend your income on video games, you cannot spend it on movies.
- **Principles of Microeconomics** - People face **trade-offs** because of scarce resources: *To get one thing, we usually have to give up something else.*
-The cost of something is what you give up to get it: *There is always an **opportunity cost** to every action.*
-Rational people think at the **margin**: *Marginal or incremental adjustments in production /revenue and costs determine economic decisions.*
-People respond to **incentives**: *There are benefits and costs of all choices, and rational agents choose actions where benefits exceed costs.*
-**Markets** are usually a good way to organise economic activity: *Resources are allocated as a result of buyers' and sellers' decisions made in their own self-interest.*
-**Government** can sometimes improve market outcomes: *As markets can **fail**.*

Branches of Economics - Microeconomics: focuses/analysis on the actions of individual agents within the economy, like households, workers, and businesses.

-Macroeconomics: looks at the economy as a whole. It focuses on broad issues such as growth of production, the number of unemployed people, the inflationary increase in prices, government deficits, and levels of exports and imports.

- **Factor of Production** - Scarce resources are typified in economics in four groupings: Scarce resources are typified in economics in four groupings: Land → rental income, Labour → wages and salaries, Capital → interest, Entrepreneurship → profit
- **Assumptions** - *efficiency*—the economy is operating at full employment and full production.
fixed resources—the quantity and quality of the factors of production which are available to a society are fixed.
fixed technology—no new technology or knowledge is introduced, and
The economy is being analysed at a particular point in time.
- The curvature of the production possibilities frontier shows that as we add more resources to capital goods, moving from left to right along the horizontal axis, the original increase in opportunity cost is fairly small, but gradually increases. Thus, the slope of the PPF is relatively flat near the vertical-axis intercept.
Conversely, as we add more resources to consumer goods, moving from bottom to top on the vertical axis, the original declines in opportunity cost are fairly large, but again gradually diminish. Thus, the slope of the PPF is relatively steep near the horizontal-axis intercept. In this way, the law of increasing opportunity cost produces the outward bending shape of the production possibilities frontier.
- A producer has an **absolute advantage** in producing a good or service if the he/she can produce that product using **fewer** resources.
A producer has a **comparative advantage in** producing something if the opportunity cost of that production is lower than other producer.
- **Demand and Supply** - Demand: the amount of some **good or service** consumers are **willing** and **able** to purchase at each **price**.
Law of demand: **Other things being constant (*ceteris paribus*)**, the higher the price of a good or service, the less consumers will demand of that good or service. The lower the price, the greater consumers will demand of that good or service.
There is an inverse relationship between the price of the good or service and the quantity demanded.
- **Why is the demand curve downward sloping?** - The substitution effect and the income effect
- **Moving vs Shifting** - A **price change** causes a movement up or down the demand curve. It is known as Qd (quantity demand).
- A change in any **non-price factor** of demand causes a shift in the demand curve, either to the right or left. This is known as shifting D (demand).
- **Non-price factors (Examples)** --1- Number of consumers in the market, --2- Income of the consumer segment
--3-Taste and preferences of the consumers --4- Expectations of the consumers, including future price expectations

Prices of related goods and services → these are substitutes and complementary goods and services.

- **Law of Supply** - Supply: the amount of some **good or service** a producer is **willing** to supply at each **price**.

Law of supply: all other things being constant (*ceteris paribus*), the higher the price of a good or service, the higher the quantity supplied of that good or service, and the lower the price, the lower the quantity supplied of that good or service.

- Supply curve shift and move is same as demand
- **Non-price factors (Examples)** - Number of suppliers or producers - The price of substitutes in production - Price of inputs - Future price - Technological advancement
- **Market Equilibrium** - In a competitive market economy, market equilibrium occurs at the interaction of demand and supply. Together, demand and supply determine the price and the quantity that will be bought and sold in a market.

- **Elasticity** - Elasticity measures the sensitivity of one variable to a change in another variable.
- Most common forms of demand elasticity are
 1. price elasticity of demand
 2. income elasticity of demand
 3. cross-price elasticity of demand.
- **Price Elasticity of Demand (PED)** - PED measures the responsiveness (sensitivity) of the quantity demanded of a particular good or service to changes in its price.

$$\text{Price Elasticity of Demand} = \% \text{ Change in Quantity Demanded} / \% \text{ Change in Price}$$

- If this formula gives a number **greater than 1**, the demand is **elastic**. In other words, quantity changes faster than price.
- If the number comes out to be **less than 1**, demand is **inelastic**. In other words, quantity changes slower than price.
- If the number is **equal to 1**, the **elasticity of demand is unitary**. In other words, quantity changes at the same rate as price.

Factors Influencing PED- 1. Nature or Type of Good, 2. Availability of Substitutes, 3. Share of Income Spent on Good, 4. Time Period

- **Income Elasticity of Demand-** Income elasticity of demand is the percentage change in the quantity demanded as a result of a 1% change in the income of the consumer.
 - A positive relationship between income and demand is a prior expectation for **normal goods**.
 - An increase in income may be associated with a fall in demand of the **inferior good**.
- **Cross-price Elasticity of Demand-** Cross-price elasticity is the % change in the quantity of good Y as a result of a one percent change in the price of good X.
 - In case of **substitute goods**, the cross-price elasticity of demand would be positive, that is, when the price of the substitute good increases, the demand for the other good will increase – as the substitute has now become relatively expensive.
 - In case of **complementary goods**, the cross-price elasticity of demand would be negative.
- **Price Elasticity of Supply-** Price elasticity of supply measures the responsiveness (sensitivity) of the quantity supplied of a particular good or service to changes in its price.
 - Price elasticity of supply shows the percentage change in the quantity supply in response to a percentage change in price.

• PED and Revenue

Price Elastic		
↑	An increase in price...	↓ reduces total revenue.
↓	A reduction in price...	↑ increases total revenue.
Total revenue moves in the direction of the quantity change.		
Price Inelastic		
↑	An increase in price...	↑ increases total revenue.
↓	A reduction in price...	↓ reduces total revenue.
Total revenue moves in the direction of the price change.		
Unit price Elastic		
↑	An increase in price...	● no change in total revenue.
↓	A reduction in price...	● no change in total revenue.
Total revenue does not change as price changes.		

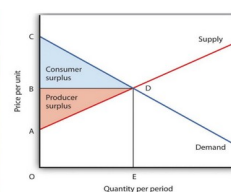
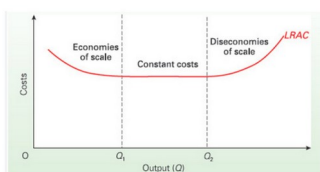
- **Factors Effecting Supply Elasticity -**
 - Number of producers: More the number of producers, greater will be the elasticity of supply.
 - Spare capacity: it is easy to increase production if there is spare production capacity available. This will make supply more elastic.
 - Ease of switching: if production of goods can be varied in the same set up, supply is more elastic.
 - Ease of storage: when goods can be stored easily, the supply response can be more elastic
 - Length of production period: quick production run time enables a quicker response to a price increase, which means more elastic supply.
 - Factor mobility: when moving resources into the industry is easier, the supply curve is more elastic.

- **Important Concepts - Productivity:** most important driver of business profitability and economic growth.
 - Business productivity can be viewed from two lenses:
 - i. producing more output using the same level of resources or ii. Producing the same level of output using fewer resources.
 - **Short-run:** a situation in which some resources are variable while others are fixed.
 - **Long-run:** a situation where firms can change the quantities of all resources employed.
 - **Production function** shows the relationship between the total output (Q) as a function inputs. Example: $Q = f(K, L)$
 - **Total product (TP):** total level of production by the firm. TP can also be represented by Q.
 - **Marginal product (MP):** additional level of output (change in output), given a change in the variable input. MP with respect to labour = $\text{Change in } Q / \text{Change in Input}$
 - **Average product (AP):** average product which is the total product divided by the level of input. AP with respect to labour = Q/L
- **Law of Diminishing Marginal Returns - 1.** When more units of a variable input are added to the production process in the **short run**, that is where there is **at least one fixed input** in the production mix, the additional output generated from the additional amount of variable input **will begin to increase**.
 - 2. After a point, if a producer continues to employ additional input, **total product will start to decline**. This will result in marginal product falling below zero and show negative returns. This is referred to as **negative marginal returns to scale**.

Accounting Vs Economic Profit - **Accounting profit** is total revenue less total explicit costs and it is positive when total revenue exceeds explicit costs.

Economic profit is total revenue less total explicit costs less total implicit costs. Economic profit can also be viewed as accounting profit less implicit costs.

- **Costs of Production-**
 - **Total fixed costs (TFC):** Fixed costs are those costs that do not vary with the quantity of output produced.
 - **Total variable costs (TVC):** Variable costs are those costs that do vary with the quantity of output produced.
 - **Average cost (AC):** Dividing the firm's costs by the quantity of output it produces.
 - **Average Total Costs (ATC):** Summation of Average Fixed Costs (AFC) and Average Variable Costs (AVC). As output increases AFC decreases.
 - **Marginal cost (MC):** Measures the increase in total cost that arises from an extra unit of production.
- **Returns to Scale/Economies of Scale**
 - Long-run average-total-cost curve will be flatter U-shape.
 - When long-run average total cost declines as output increases, there are said to be economies of scale.
 - When long-run average total cost rises as output increases, there are said to be diseconomies of scale.
 - When long-run average total cost does not vary with the level of output, there are said to be constant returns to scale.
- **Economies of Scope**
- Economies of scope reduce average costs when firms make multiple products.
- They are typically observed in the joint production of different variants of the same basic product.
- Economies of scope occur whenever inputs can be shared in the production of different products.

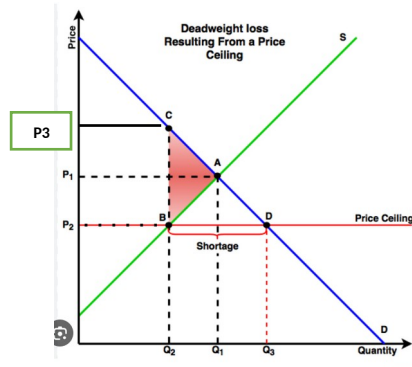
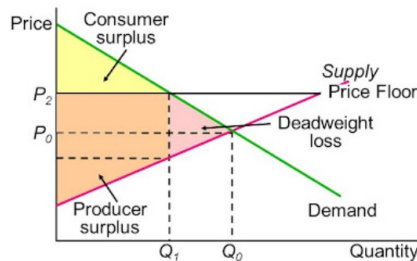


- **Consumer Surplus and Producer Surplus**
- **Market Failure**
 - Markets can fail to equilibrate at a level of production that maximises resource use efficiency.
 - Markets can get its pricing incorrect.
- **Example of Market Failure**
- Price Control
 - Ceiling price
 - Floor price
- Externality
- Lack of competition in the market
- Public goods
- **Price Control: Ceiling Price**
 - When government thinks that market price is too high and wishes to decrease the price to make it more affordable for consumers, it may place a binding price ceiling of that particular market.
 - This policy means that sellers cannot charge more than the ceiling price.
 - This could create shortage in the market as price is cheaper than the equilibrium price.

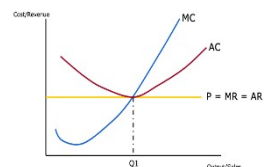
- It leads to market failure, causing deadweight loss.
- **Price Control: Floor Price**
 - When government increases the price above the equilibrium price to support the producers is known as a binding floor price.
 - As price is now higher than the equilibrium, many consumers will reduce their consumption and that would create surpluses in the market.
 - This will also lead to market inefficiency with deadweight loss.

• Dead Weight loss (DWL)

Effect of Price Floor



- **Externality**
 - An externality is the impact that a firm's or individual's action has on a bystander.
 - An externality is a cost or benefit imposed on third parties (people not directly involved in the buying and selling of that good).
 - As a result of externalities, the socially optimal output differs from the output that a firm might generate in order to maximise profits.
- **Types of Externalities**
 - Positive externality (Example: government expenditure in schooling, Vaccination)
 - Negative externality (Example: loud music, noise pollution caused by aircraft, smoke from a factory)
- **Policies for Externality**
 - Policies to correct negative externality - Taxation & Regulation
 - Policies for positive externality - Incentives and Subsidy
- **Lack of Competition**
 - When firms producing a product collude and engage in anti-competitive behaviour.
 - It reduces economic surplus and market outcomes that are not good for consumers and society in general.
 - In Australia, Australian Competition and Consumer Commission (ACCC) regulate and encourage firms not to exert market power to the detriment of consumers and society.
- **Public Good**
 - **Public good:** When a good is both **non-excludable** and **non-rival** in consumption. Example: National defence, Street lighting.
 - The benefits of these goods are such that exclusion is not feasible. Once they are produced, anyone can enjoy them; there is no practical way to exclude people who have not paid for them from consuming them.
 - **Private good:** When a good is both **excludable** and **rival** in consumption. Example: mobile phone, pizza
- **Free-rider Problem**
 - Public goods suffer from Free-rider problem.
 - **Free riders** are people or firms that consume a public good without paying for it. Taxpayers finance public goods.
 - When too many free riders consume a shared resource, it becomes difficult to maintain that resource, which is known as the free rider problem.
 - Public good suffers from inefficiently low production due to free-riders.
- **Most known market structures in Economics are:**
 - Perfect competition/perfect competition/ perfectly competitive market/competitive market/ Perfect market (PM)
 - Monopoly
 - Monopolistic competition
 - Oligopoly
- **PERFECT COMPETITION (below are the characteristics)**
 1. There are large numbers of sellers and buyers.
 2. No firm has **market power** over others and therefore all firms are purely **price takers**.
 3. All the firms produce identical (or homogenous) products.
 4. There are no barriers to entry and exit.
 5. All sellers and buyers possess perfect knowledge.
 6. $P = AR = MR = MC$ (6th one)
- **Profit Maximization Condition in PC** - better to stop producing additional products when MC intersects with AC, because after which, to produce each additional product cost involved is high with less gain.
 - $P = AR = MR = MC$



- When, $MR > MC$, total profit is increasing
- When $MC > MR$, every unit's sale after this point reduces total profits
- When $MC = MR$, profit will be maximised

• Shut down and Exit

- A **shutdown** refers to a **short-run** decision not to produce anything during a specific period because of current market conditions.
- firm should shut down if the revenue it would earn from producing is less than the variable costs of production. $TR < VC$
- An **exit** refers to a **long-run** decision to leave the market.
- firm should exit if the revenue it would earn from producing is less than its total costs. $TR < TC$

• Efficiency in PC

Perfectly competitive market allows productive and allocative efficiencies.

1. Productive efficiency is where the highest level of output is produced with the **least resources**. This happens when a firm is operating at the bottom of its average total cost (ATC) curve.
2. Allocative efficiency is achieved when the value consumers place on the good or service equals the cost of the resources used in production. This happens when the price is equal to its marginal cost ($P=MC$).

• MONOPOLY (below are the characteristics)

- Only one producer with no competition in the market.
- There is only one product for which there are no close substitutes.
- Extremely high barriers to entry.
- Minimal advertising expenditure.
- Price Maker.

• Sources of monopoly market power

- The government may be blocking the entry of more than one. For example, pharmaceutical companies producing COVID-19.
- One firm has control of a key raw material necessary to produce a good. For instance, De Beers enjoyed a monopoly on the world's diamond market.
- Economies of scale: so large that one firm has a natural monopoly. For example, more than one corner store in the same street may make the business non-profitable for both, so the second operator would generally not set up shop.
- Pricing strategies: As a price maker, a monopoly can erect a price barrier to prevent a new player from entering the market.
- Market Franchise: This is a contract entered into by a governmental organisation and a business firm providing the firm with an exclusive right to market a good or services within its jurisdiction. For example, VFS is an immigration firm.

• Price Discrimination

- It is a pricing technique that a monopolist can use to maintain a super-normal profit.
- It is the practice of a seller charging different customers different prices for the same product not justified by cost differences.
- It allows sellers to charge what buyers are willing to pay.
- It is not possible in perfect competition, but possible in any market where firms enjoy some form of market power.

• Conditions of Price Discrimination

- A firm can engage in price discrimination if:
 - the seller is **a price maker**
 - the seller can segregate consumers into different market segments (perhaps on the basis of price elasticity of demand)
 - the seller can prevent arbitrage (buying then reselling for a higher price). That is, customers who are segregated to lower price segments cannot resell to customers in high price segments – therefore there is no third-party selling. (airline tickets)
 - the practice of price discrimination does not cause resentment among customers in the higher price segments.
 - the act of price discrimination is not illegal.

Types of Price Discrimination

- First-degree price discrimination - Firm charge a different price to each customer along the demand curve. This will pretty much extract the entire consumer surplus. For example, buying a house from a developer via independent auctions for each house.
- Second-degree price discrimination - This is also called indirect price discrimination. It involves the firm charging different prices based on the volume consumed. For example, discount for higher quantity consumers.
- Third-degree price discrimination - This involves the firm segmenting the market into certain groups and then charging a different price to each group. For example, software discounts to student users, rail way, air lines

MONOPOLISTIC COMPETITION (below are the characteristics)

- There are many small firms
- Products are differentiated (unique and not homogenous)
- Low barriers to entry and exit
- Existence of non-price competition

- **Examples** – Restaurants / Private schools / Health clubs / Hairdressers / clothing shops
- **Efficiency**
 - Since monopolistic firms manipulate price by restricting output, there will be loss of consumer and producer surplus.
 - A monopolistic market in most cases is more efficient than a monopoly.
 - Economists have also argued that the product variety and consumer choice that occurs because of product differentiation may outweigh efficiency loss.
- **OLIGOPOLY (below are the characteristics)**
 - Few large sellers
 - Homogeneous or differentiated goods and services
 - Actions of a single firm will potentially have an impact on the profits of its rivals.
 - Mutual Interdependence: Interdependence between firms gives rise to *strategic behaviour*. Firms must make decisions keeping in mind the actions of other firms within the industry.
 - Significant entry barriers
 - Significant non-price competition
- **Examples** - Banking Ind, Telecommunication Ind, Automobile Ind, insurance comp, Media ind, pharmaceutical companies
- **Models of oligopoly**- The five main models are:
 1. Non - price Competition
 2. Price Leadership
 3. The Cartel
 4. The Kinked Demand Curve
 5. Game Theory
- **Non-price competition**
 - Major oligopolies often use non-price techniques such as advertising or product differentiation as a source of competitive advantage.
 - They also use techniques such as research and development (R&D) to develop new products that are difficult for others to imitate.
- **Price Leadership**- This is a pricing strategy in which a dominant firm sets the price for an industry and the other firms follow. It should be noted that the dominant firm is not necessarily the largest.
- **Cartel**
 - A cartel is a group of firms formally agreeing to control the price and output of a product from cooperation and self-interest.
 - In the case of cooperation, the oligopolists may put up a united front and act like a monopolist by producing a lower output and charging a price higher than marginal cost to earn super normal profits in both the short-run and long-run.
 - Cooperation often breaks down either through (a) one or more firms reneging and acting in self-interest or, (b) by the hand of law. Creating a situation like a monopoly is illegal under the Trade Practice Act (1974) and penalties include both fines and jail time for Directors and the senior management.
- **Example:** Organisation of the Petroleum Exporting Countries (OPEC) and International Air Transport Association (IATA).
- Cartels may control markets but there is an incentive for members to “cheat”. For example, during the 2022 Russian- Ukrainian situation, several members of the OPEC broke ranks and acted in self-interest independently of each other.
- **Game Theory: Prisoners’ Dilemma**
 - Prisoners’ Dilemma is a game where pursuing dominant strategies results in non-cooperation that leaves everyone worse off.
 - A dominant strategy is one that is the best for a firm, irrespective of what strategies other firms use.
 - This game shows why sustained cooperation between competitors may not be feasible. It can be observed in business and indeed in life, people fail to cooperate with one another even where cooperation would make them all better off.
- difference of micro and macro economics
- **Microeconomics is the study of individuals and business decisions, while macroeconomics looks at the decisions of countries and governments.** Though these two branches of economics appear to be different, they are interdependent and complement one another. Many overlapping issues exist between the two fields.